

Functional Analysis — Homework 2

Connor Mooneyhan

Problem 1. Recall that c_0 is the space of sequences converging to zero, equipped with the supremum norm: $\|x\|_\infty = \sup_j |x_j|$. Consider the subspace:

$$S = \left\{ (x_1, x_2, x_3, \dots) : \sum_{j=1}^{\infty} \frac{x_j}{2^j} = 0 \right\}.$$

Show that there is no unit vector $x \in c_0$ with $d(x, S) = 1$. Hint: start with an arbitrary unit vector $x \in c_0$, and try to construct a vector $y \in S$ such that the distance $\|x - y\|_\infty$ is less than 1.

Proof. The best idea I've had for this is to take $x = (x_1, x_2, x_3, \dots) \in c_0$ and derive the sequence $x' = (x_1/2, -x_2/4, x_3/8, \dots)$ from it. I'm not sure how to show that the resulting sum $\sum_{j=1}^{\infty} (-1)^{j+1} x_j / 2^{2j}$ converges to 0, though, if indeed it does (I'm leaning towards thinking it doesn't, based on pairing up terms in the partial sums). The partial sums don't seem to play nicely enough to work with directly, and I don't recall any "tests" or formulas from Calc 2 that would help. $\square$

Problem 2. Let V be a normed space, and $x \neq y$ elements of V . Show that there exists a bounded linear functional $f \in V^*$ such that $f(x) \neq f(y)$.

Proof. Define a linear functional f on $\text{span}(x, y)$ by $x \mapsto 1$ and $y \mapsto 2$. Since $\text{span}(x, y)$ is finite-dimensional and every linear map from a finite-dimensional space is continuous, we have that f is continuous and thus bounded. By Hahn-Banach, we can extend this to a bounded linear functional $\tilde{f}$ on V , and we are done. $\square$

Problem 3. Let V be a normed space, V^* its dual and V^{**} its bidual. Consider the map

$$j : V \rightarrow V^{**}$$

given by $j(v)(f) = f(v)$.

- (a) Show that j is well-defined, i.e. $j(v)$ is a bounded linear functional on V^* .
- (b) Show that j is linear.
- (c) Show that j is an isometry onto its image.

This problem shows that V embeds isometrically into a subspace of V^{**} under j . In general, j need not be surjective. When it is, we say that V is reflexive.

Proof.

(a) For linearity of $j(v)$,

$$\begin{aligned} j(v)(af + bg) &= (af + bg)(v) \\ &= af(v) + bg(v) \\ &= aj(v)(f) + bj(v)(g). \end{aligned}$$

Now assume that $\|f\|_{op} = 1$. Then

$$\begin{aligned} \|j(v)(f)\| &= \|f(v)\| \\ &\leq \|f\|_{op}\|v\| \\ &= \|v\|, \end{aligned}$$

so since $\|v\|$ is constant as we vary f , $j(v)$ is bounded.

(b) We have

$$\begin{aligned} j(av + bx)(f) &= f(av + bw) \\ &= af(v) + bf(w) \\ &= aj(v)(f) + bj(w)(f) \\ &= (aj(v) + bj(w))(f), \end{aligned}$$

where the second equality comes from the linearity of f .

(c) We already saw in the proof of part (a) that $\|j(v)(f)\| \leq \|v\|$ whenever $\|f\|_{op} = 1$. Thus $\|j(v)\|_{op} \leq \|v\|$, so if we can find an f with $\|f\|_{op} = 1$ such that $\|j(v)(f)\| = \|v\|$, we're done since we will have demonstrated that the supremum of $\|v\|$ is achieved. We can define a bounded linear map $f : \mathbb{R}v \rightarrow F$ by choosing to map $v \mapsto \|v\|$, and then by Hahn-Banach we can extend this to a bounded linear map $\tilde{f} : V \rightarrow F$ so that $\tilde{f} \in V^*$. Now we just check that indeed

$$\|j(v)(\tilde{f})\| = \|f(v)\| = \|v\|$$

as desired, and we are done. □

Problem 4. Let V be a normed space and

$$S_1(V) = \{v \in V : \|v\| = 1\}$$

be its unit sphere. Prove that V is separable if and only if $S_1(V)$ is separable.

Proof. If V is separable with countable dense subset D , let $D_1 = D \cap S_1(V)$ and let $x \in S_1(V)$. Then given $\varepsilon > 0$, $B_\varepsilon(x) \cap D \neq \emptyset$, so we can choose $y \in B_\varepsilon(x) \cap D$ such that $y \geq 1$ (if not, we can run a similar argument for $y \leq 1$), and we have

$$\begin{aligned} 1 \leq \|y\| &\leq \|y - x\| + \|x\| \\ &= \|y - x\| + 1 \\ &< \varepsilon + 1, \end{aligned}$$

so we can find elements of D arbitrarily close to the unit sphere that are within ε of x , but not necessarily *on* the unit sphere. I tried taking $y/||y||$ and seeing if it was closer to x than y was. I thought it might be, but my efforts to prove it yielded nothing useful.

For the other direction, if $S_1(V)$ has a countable dense subset D , then for any $x \in V$, we have $(x_n) \rightarrow x/||x||$ where $x_n \in D$ for every n . I hoped to use these sorts of sequences to show that something like, say, $\{qd : q \in \mathbb{Q}, d \in D\}$ was a countable dense subset of V , but had no luck in trying to make that argument work. $\square$

Problem 5. (Uniform Boundedness Principle). Let X be a Banach space, Y a normed space, and $\{T_\alpha\}_{\alpha \in J}$ an arbitrary family of bounded linear operators

$$T_\alpha : X \rightarrow Y$$

satisfying the **pointwise bound**:

$$\text{For each } x \in X, \quad \sup_{\alpha \in J} ||T_\alpha(x)||_Y < \infty.$$

Use the Baire Category Theorem to show that this implies the **uniform bound**:

$$\sup_{\alpha \in J} ||T_\alpha||_{op} < \infty.$$

Proof. Really not sure what to do here. For the Baire Category Theorem, I need a collection of closed sets with no interior, so my only real idea for that was taking preimages of certain closed sets, but I haven't been able to figure out what closed sets I would use (singletons don't seem to obviously give me empty interior). I also tried to work it backwards, thinking "how could the conclusion that my closed sets don't union to the whole space help me?" The only thing that seems remotely plausible to me is to somehow have my closed sets be in $\mathbb{R}$ and construct some sort of correspondence so that the size of their union is $\sup_{\alpha \in J} ||T_\alpha||_{op}$, and then show that their union is finite... but I have no idea how to move forward with it, and I definitely don't feel like that's actually the way to do it. $\square$